

Algorithmic Feed Simulation Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	algorithmic-feed-simulation

Big idea: A feed is a knowledge filter.

Student-Facing Prompt

Inspect Algorithmic Feed Simulation Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Algorithmic Feed Simulation Stimulus A	Student-facing stimulus 1: Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Algorithmic Feed Simulation Stimulus B	Student-facing stimulus 2: A feed that changes after each student 'likes' or skips a card. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Algorithmic Feed Simulation Reveal / Twist	Student-facing stimulus 3: A comparison between a curiosity-maximising feed and a reliability-maximising feed. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A feed is a knowledge filter.

Actual Lesson Questions

#	Question for students
1	After inspecting Algorithmic Feed Simulation Stimulus A, what is your first knowledge claim?
2	Which exact detail from Algorithmic Feed Simulation Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Do algorithms expand knowledge or narrow it?

Ready-to-Use Examples

- Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks.
- A feed that changes after each student 'likes' or skips a card.
- A comparison between a curiosity-maximising feed and a reliability-maximising feed.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Activity Steps

1. Students secretly choose five cards they would click.
2. The teacher acts as the algorithm and gives each student more cards similar to their choices.
3. Repeat for three rounds.
4. Students compare how different feeds became over time.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Do algorithms expand knowledge or narrow it?
- Is a personalised feed a window, a mirror or a cage?
- Who is responsible for algorithmic bias: user, designer, company or society?

Knowledge Questions

- Do algorithms expand knowledge or narrow it?
- Is a personalised feed a window, a mirror or a cage?
- Who is responsible for algorithmic bias: user, designer, company or society?

Transfer Example

For example, run this activity with paper post cards ranked by engagement, recency, outrage, or similarity to past clicks. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...